

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes

Total Marks 20

Annexure A

Descriptive Written Test

Code of Conduct:

I V-Surya Satvika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V-surya satvika

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20

B

CO1-AT-3
Assessment tool-3

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A Smart home automation system is being designed to manage lighting, improves comfort, security and energy efficiency by automatically controlling devices like lights, air conditioners and sensors such as motion, temperatures and time.

- ① A simple reflex agent
- ② Model-based agents
- ③ Goal-Based agents
- ④ utility-based agent.

Are the Intelligent Agents. *

A Simple reflex agent:-

This works on fixed rules based only on current inputs. For example, it turns on light when motion is detected or activates an alarm when door opens at night. It is fast memory but has no memory or learning ability.

Model-Based agent:-

This stores past information and uses it to make better decision. For instance, it may delay turning off lights after someone leaves a room or adjust temperature based on daily routines.

Goal-Based Agent:-

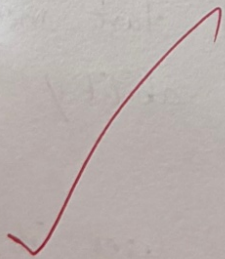
This acts to achieve specific goals like maintaining comfort, saving energy or ensuring security. It plans action before choosing the best option.

Utility-Based Agent:-

This selects actions that given the best overall benefit by balancing comfort, cost, energy use and safety. For example, It may cool only occupied rooms to save electricity.

A hybrid system combining these agents is most effective, as it provides quick responses, memory, planning and optional decision-making.

A hybrid Intelligent agent is best for smart homes because it improves comfort, security and efficient energy while adapting to user needs.



A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. uninformed search algorithms like Uniform cost search (UCS) and Iterative Deepening search (IDS) help the robot explore the maze and reach the goal.

① Uniform cost search (UCS) Algorithm

This always selects the levels by levels increasing depth limit. It uses less memory and works well the goal is reached.

② Iterative Deepening Search (IDS) Algorithm

IDS searches level by level depth limit. Unknown but it repeats the search at each level.

③ Intelligent Agents

A model-Based Agents remembers previously explored paths, while a Goal-Based Agent plans actions to reach the exit. A simple Reflex Agent is less suitable because it cannot remember past information.

④ Best Approach

A Goal-Based Agent with UCS is the best is the best combination because it focuses on reaching the goal while finding least-cost path.

Goal-Based agent with IDS is good alternative.

Conclusion

combining a Goal-Based with an appropriate search algorithm enables the robots to navigate the maze efficiently and successfully which the exit.